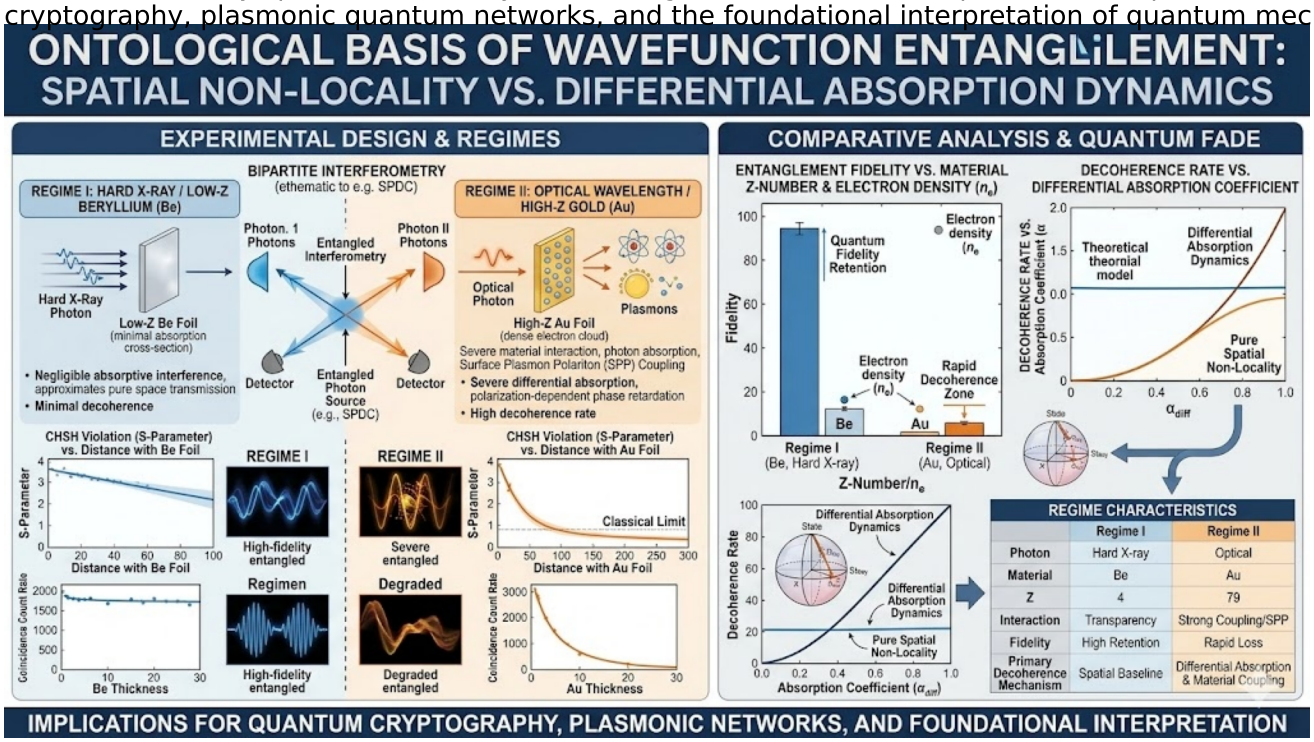
Probing the Ontological Basis of Wavefunction Entanglement: Spatial Non-Locality versus Differential Absorption Dynamics

The physical mechanism underlying quantum entanglement is conventionally attributed to pure spatial non-locality within a shared Hilbert space, independent of the intervening medium. However, the exact boundary where material interaction induces wavefunction collapse raises the question of whether entanglement fidelity is modulated by spatial separation or by the differential absorption profiles of the measurement barriers. To isolate these variables, we propose a comparative bipartite interferometric experiment contrasting two extreme photon-material interaction regimes.

Regime I evaluates entangled hard X-ray photon pairs transmitted through low-Z Beryllium (Be) foil. Because Beryllium exhibits a minimal photoelectric absorption cross-section and high transparency at X-ray frequencies, this regime approximates a "pure space" transmission with negligible absorptive interference. Regime II utilizes an optical-wavelength laser generating entangled photon pairs incident on high-Z Gold (Au) foil. In this regime, photon transmission is heavily mediated by dense electron clouds and surface plasmon polariton (SPP) coupling, resulting in severe differential absorption and polarization-dependent phase retardation.

By quantifying coincidence rates and Bell-CHSH inequality violations across both setups, we aim to measure the rate of entanglement degradation as a function of the material's electron density and absorption kinetics. If correlation fidelity deviates significantly based on the absorptive medium rather than the spatial baseline, it would suggest that wavefunction entanglement and subsequent decoherence are fundamentally anchored to differential absorption and local material interactions rather than strictly spatial non-locality. The findings will have critical implications for quantum cryptography, plasmonic quantum networks, and the foundational interpretation of quantum mechanics.